
DRP Final Presentation

Started with Decision Theory, ended with Model Selection

The True Model Perspective

There exists an underlying data-generating distribution that 'explains' variables and randomness.

“This approach takes very seriously the “some models are useful” part of Box’s aphorism, but it ignores the “all models are wrong” part”

Model Selection (single model view)

The standard practice of introductory statistics classes, you select a single model or family of models that is believed to best represent the observed data. The selected model is assumed to have generated the observations.

Bayesian Model Averaging

Posterior distribution is the “average of the posterior distributions under each of the models considered, weighted by their posterior model probability” (Hoeting, 1999)

$$\text{pr}(\Delta | D) = \sum_{k=1}^K \text{pr}(\Delta | M_k, D) \text{pr}(M_k | D).$$

$$\text{pr}(M_k | D) = \frac{\text{pr}(D | M_k) \text{pr}(M_k)}{\sum_{l=1}^K \text{pr}(D | M_l) \text{pr}(M_l)}$$

$$\text{pr}(D | M_k) = \int \text{pr}(D | \theta_k, M_k) \text{pr}(\theta_k | M_k) d\theta_k$$

Some issues (with resolutions):

- Massive summation (too many models)
 - Difficult integral for the posterior model probability
 - Challenging to specify the prior
 - Choosing the class of models to average
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Bayesian Model Averaging

Resolving the aforementioned issues:

- Too many models:
 - Occam's Window: Exclude complex or worse performing models
 - Difficult integral:
 - Laplace and BIC approximations
 - MLE approximation
 - Specifying the priors:
 - Start with a uniform, update using 'imaginary data' provided by a domain expert (Madigan & Raftery, 1995)
 - Choosing model class:
 - Include only scientifically plausible models
 - Explore all models consistent with the data
 - Start broad and prune the model class (Draper, 1995)
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Proper Scoring Rules

A proper scoring rule, according to Parmigiani & Inoue, is one whose expected score is maximized by the true predictive distribution (one that best represents the forecaster's 'true belief')

Examples include:

- Logarithmic score
 - Brier score
 - CRPS
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Models in Decision-Making(Höppner, 2026)

Compares dominant frameworks for modeling decision making under uncertainty.

“the two pillars of EUT—[stable preferences and accurate judgments]—are context-sensitive, cognitively costly, and socially contingent.”

Many models are proposed to resolve these issues, such as but not limited to Modern EUT, Heuristics, Regret Theory, Risk-Sensitivity Theory, and more.

Moving Forward

- Finish reading Decision Theory by Parmigiani & Inoue
 - Finish reading Höppner's paper
 - Move from conceptual to technical
 - The open model perspective
 - Model class is defined iteratively as the analysis goes on
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